

Question 1 : What is Simple Linear Regression (SLR)? Explain its purpose.

Answer : Simple Linear Regression (SLR) is a **statistical and machine learning technique** used to understand and model the **linear relationship between one independent variable (X) and one dependent variable (Y)**. It assumes that the dependent variable changes at a constant rate with respect to the independent variable.

The relationship between X and Y is represented by a straight-line equation:

$$Y = b_0 + b_1X$$

where:

- b_0 is the intercept
- b_1 is the slope of the line

Purpose of Simple Linear Regression:

- To **predict** the value of the dependent variable based on the independent variable
- To **analyze the strength and direction** of the relationship between variables
- To identify **trends and patterns** in data
- To help in **decision-making and forecasting**

Question 2: What are the key assumptions of Simple Linear Regression?

Answer : The key assumptions of Simple Linear Regression are:

1. **Linearity:** A linear relationship exists between X and Y.
2. **Independence:** Observations are independent of each other.
3. **Homoscedasticity:** Constant variance of errors across all values of X.
4. **Normality:** Residuals are normally distributed.
5. **No Autocorrelation:** Error terms are not correlated.

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Answer : The mathematical equation of a Simple Linear Regression model is:

$$Y = b_0 + b_1X$$

Where:

- **YYY** is the **dependent variable** (output)
- **XXX** is the **independent variable** (input)
- **b₀** is the **intercept**, representing the value of Y when X = 0
- **b₁** is the **slope**, representing the change in Y for a one-unit change in X

This equation represents a **straight-line relationship** between the independent and dependent variables.

Question 4: Provide a real-world example where simple linear regression can be applied.

Answer : Simple Linear Regression can be applied to predict house prices based on area.

In this case, the independent variable (X) is the size of the house (in square feet), and the dependent variable (Y) is the house price. The model helps estimate how the price changes with an increase in house area.

Question 5: What is the method of least squares in linear regression?

Answer : The method of least squares is a technique used to estimate the parameters of a linear regression model by minimizing the sum of the squared differences between the observed values and the predicted values.

It finds the best-fit line by ensuring that the total squared error between actual data points and the regression line is as small as possible.

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Answer : Logistic Regression is a supervised machine learning algorithm used for classification, especially binary classification problems. It predicts the probability of an outcome using the sigmoid function, which gives values between 0 and 1.

Logistic Regression differs from Linear Regression in several ways. Logistic Regression is used for classification, while Linear Regression is used for predicting continuous values. Logistic Regression produces probability-based outputs, whereas Linear Regression produces numerical outputs. Logistic Regression applies a sigmoid function, while Linear Regression uses a straight-line equation.

Question 7: Name and briefly describe three common evaluation metrics for regression models.

Answer : Three common evaluation metrics for regression models are:

1. Mean Absolute Error (MAE):

Measures the average absolute difference between actual and predicted values.

2. Mean Squared Error (MSE):

Measures the average of the squared differences between actual and predicted values, giving more weight to larger errors.

3. Root Mean Squared Error (RMSE):

Measures the square root of MSE, providing error values in the same units as the target variable.

Question 8: What is the purpose of the R-squared metric in regression analysis?

Answer : The **R-squared (R^2) metric** is used to measure how well a regression model explains the **variation in the dependent variable**. It represents the **proportion of variance** in the output that is explained by the input variables.

An R-squared value closer to **1** indicates a **better model fit**, while a value closer to **0** indicates a poor fit.

Question 9: Write Python code to fit a simple linear regression model using scikit-learn and print the slope and intercept. (Include your Python code and output in the code box below.)

Answer :

```
from sklearn.linear_model import LinearRegression
import numpy as np
```

```
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
y = np.array([2, 4, 6, 8, 10])
```

```
model = LinearRegression()
model.fit(X, y)
```

```
print(model.coef_[0])
print(model.intercept_)
```

Output

2.0

0.0

Question 10: How do you interpret the coefficients in a simple linear regression model?

Answer : In a Simple Linear Regression model, the coefficients explain the relationship between the independent and dependent variables. The intercept (b_0) represents the value of the dependent variable when the independent variable is zero. The slope (b_1) represents the change in the dependent variable for every one-unit increase in the independent variable.

Together, these coefficients describe how the dependent variable responds to changes in the independent variable.